TEEL TAAL

A proof of Roth's theorem

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How large can a set of integers be without containing three equally spaced distinct elements? These notes give a self-contained Fourier-analytic proof that a set with no such triple must have density zero.

We begin with arithmetic progressions and their history. We then place an integer interval inside a finite cyclic group, develop its Fourier theory, and prove three results: a large Fourier coefficient, a density increment on an integer progression, and a finite iteration. These three taals combine to prove Roth's theorem.

The proof uses finite sums, elementary group theory, Cauchy–Schwarz, and the pigeonhole principle. The Fourier identities and the approximation, partition, and iteration lemmas are all proved below. Historical results are stated for context; the proof of Roth's theorem does not use them.

1.1 Notation: arithmetic progressions

Write $\mathbb{N} = \{1, 2, \dots\}$ and $[N] = \{1, \dots, N\}$.

Definition 1.1. For an integer $k \geq 2$, a *k-term arithmetic progression* (k-AP) in $\mathbb{Z}$ is a sequence

$$x,\ x + d,\ \dots,\ x + (k-1)d.$$

It is *nontrivial* if $d \neq 0$. A set is *k-AP-free* if it contains no nontrivial k-AP.

For example, $3, 8, 13$ is a nontrivial 3-AP. For three integers $a < b < c$, equal spacing is equivalent to

$$a + c = 2b. \tag{1.1}$$

The additive equation, rather than just the picture of equal gaps, will be what makes Fourier analysis useful. Write $[N] = \{1, \dots, N\}$.

1.2 History: from colouring to density

1.2.1 Van der Waerden's theorem (1927)

Theorem 1.2 (Van der Waerden). *For every pair of integers $q \geq 1$, $k \geq 2$, there exists $W = W(q, k)$ such that every colouring of $[W]$ with q colours contains a nontrivial monochromatic k-term arithmetic progression.*

Equivalently, every finite colouring of $\mathbb{N}$ has monochromatic arithmetic progressions of every finite length. The theorem guarantees a progression in *some* colour class; it does not say that every infinite class has one.

Example. Can you exhibit an infinite set with no nontrivial three-term arithmetic progression?

The set $E = \{1, 2, 4, 8, \ldots\}$ works. If $2^a < 2^b < 2^c$ were equally spaced, then dividing $2^a + 2^c = 2^{b+1}$ by 2^a would give

$$1 + 2^{c-a} = 2^{b+1-a},$$

which equates an odd integer with an even one. Thus E is 3-AP-free, and consequently k-AP-free for every $k \geq 3$. This is an example of a *set*, not a colouring in which every colour class avoids progressions.

1.2.2 How thin is this example?

Definition 1.3. For $A \subseteq \mathbb{N}$, its *upper density* is

$$\overline{d}(A) = \limsup_{N \to \infty} \frac{|A \cap [N]|}{N}.$$

If the ordinary limit exists, it is the *natural density* $d(A)$.

For the powers of two,

$$|E \cap [N]| = \lfloor \log_2 N \rfloor + 1, \qquad \frac{|E \cap [N]|}{N} \longrightarrow 0.$$

Hence $d(E) = \overline{d}(E) = 0$. More generally, upper density zero implies that natural density exists and is zero: the ratios are nonnegative, so their liminf and limsup must both be zero.

The question is whether every progression-free set must be this thin in the sense of having density zero. For a finite formulation, define

$$r_k(N) = \max\{|A| : A \subseteq [N],\ A \text{ is } k\text{-AP-free}\}.$$

For each fixed $k \geq 3$, the density question asks whether $r_k(N)/N \to 0$.

1.2.3 Erdős–Turán, Szemerédi, and the reciprocal question

Erdős and Turán proposed the density conjecture in 1936 [2]. Szemerédi's 1975 theorem settled it for every fixed progression length [3]:

Theorem 1.4 (Szemerédi)**.** *For every integer $k \geq 3$ and every $\varepsilon > 0$, there exists $N_0(k, \varepsilon)$ such that any $A \subseteq [N]$ with $N \geq N_0(k, \varepsilon)$ and $|A| \geq \varepsilon N$ contains a nontrivial k-term arithmetic progression.*

Thus $r_k(N) = o(N)$. In particular, an infinite set of positive upper density contains progressions of every finite length: choose arbitrarily large N for which its density in $[N]$ exceeds a fixed positive number.

The primes are another useful example of a zero-density set. They are not progression-free: Green and Tao proved that the primes contain arbitrarily long arithmetic progressions [4]. Their 2004 preprint appeared in print in 2008. Szemerédi's theorem by itself cannot yield this conclusion, because its hypothesis is positive density among the integers.

Erdős also asked a stronger question:

$$\sum_{a \in A} \frac{1}{a} = \infty \quad \Longrightarrow \quad A \text{ contains arithmetic progressions of every finite length?}$$

The three-term version is recorded in his seminar of 21 May 1973 [5]; the full version is documented in the mid-1970s [6]. Positive upper density implies divergence of the reciprocal sum, but the converse fails: there are zero-density sets with divergent reciprocal sum. For

example, $a_n = \lceil n \log(n+1) \rceil$ is strictly increasing, $a_n/n \to \infty$, and $\sum_n 1/a_n$ diverges by comparison with $\sum_{n \geq 2} 1/(n \log n)$. The reciprocal question concerns sets beyond the positive-density setting.

2026 update. OpenAI's release of 6 October 2026 includes a preprint, *Quasipolynomial Bounds for Arithmetic Progressions*, announcing a proof of the full reciprocal-sum conjecture [9, 10].

1.2.4 Roth's first breakthrough (1953)

The first major density result was the case $k = 3$, proved by Roth in 1953 [1].

Theorem 1.5 (Roth)**.** *For every $\varepsilon > 0$, there exists $N_0(\varepsilon)$ such that every $A \subseteq [N]$ with $N \geq N_0(\varepsilon)$ and $|A| \geq \varepsilon N$ contains a nontrivial three-term arithmetic progression. Equivalently, $r_3(N) = o(N)$.*

Roth was 27 when his paper appeared in 1953; he received the Fields Medal in 1958 [8]. Our aim is to understand the Fourier-analytic route to Theorem 1.5. Three taals, one theorem.

1.3 Preparing the stage

1.3.1 Step I: reduction modulo $m = 2N + 1$

We wish to use periodic waves, so we place $[N]$ inside the cyclic group

$$G = \mathbb{Z}/m\mathbb{Z}, \qquad m = 2N + 1.$$

Let $\pi : \mathbb{Z} \to G$ be reduction modulo m, and let $\iota = \pi|_{[N]}$. Thus $\iota(x) = \overline{x}$, and $\iota : [N] \hookrightarrow G$ is injective. Keep the integer sets and their images distinct by writing

$$S = \iota([N]) \subseteq G, \qquad B = \iota(A) \subseteq S.$$

We have $|S| = N$ and $|B| = |A|$. More importantly, reduction preserves exactly the three-term progression equation.

Lemma 1.6 (No spurious three-term progressions)**.** *For $x, y, z \in [N]$,*

$$\overline{x} + \overline{z} = 2\overline{y} \text{ in } G \quad \Longleftrightarrow \quad x + z = 2y \text{ in } \mathbb{Z}.$$

Consequently, $A \subseteq [N]$ is 3-AP-free in the integers if and only if $B = \iota(A)$ is 3-AP-free in G.

Proof. The forward congruence says that $m \mid x + z - 2y$. But

$$|x + z - 2y| \leq 2N - 2 < m,$$

so this multiple of m must be zero. The converse is immediate by reduction modulo m. Distinct integers in $[N]$ remain distinct modulo m, so nontriviality is preserved as well. □

In an odd-order cyclic group, a progression $x, x + d, x + 2d$ with $d \neq 0$ has three distinct terms: equality of the endpoints would give $2d = 0$, and multiplication by 2 is invertible modulo m.

1.3.2 Step II: indicator functions and progression counts

Definition 1.7. For $E \subseteq G$, its *indicator function* on G is

$$\mathbf{1}_E : G \to \{0,1\}, \qquad \mathbf{1}_E(x) = \begin{cases} 1, & x \in E, \\ 0, & x \notin E. \end{cases}$$

For any function on G, normalized averaging means $\mathbb{E}_{x\in G} f(x) = m^{-1} \sum_{x\in G} f(x)$.

Throughout the Fourier calculations, $A \subseteq [N]$ remains a set of integers, while $B \subseteq S \subseteq G$ denotes its image. In particular, $\mathbf{1}_B$ and $\mathbf{1}_S$ are functions on G. There are two densities, with the same cardinality in the numerator:

$$\Delta = \frac{|A|}{N}, \qquad \delta = \frac{|B|}{m} = \frac{N}{m}\Delta. \tag{1.2}$$

Here Δ is the actual density of the chosen A in $[N]$ and depends on A, N; δ is the mean of $\mathbf{1}_B$ on G. Roth's theorem fixes a positive lower bound for the actual density as N varies.

Define the trilinear progression-counting functional

$$\Lambda_3(f_1, f_2, f_3) = \mathbb{E}_{x,d\in G} f_1(x) f_2(x+d) f_3(x+2d). \tag{1.3}$$

The product of three indicators equals one precisely when all three entries belong to B. Thus

$$m^2 \Lambda_3(\mathbf{1}_B, \mathbf{1}_B, \mathbf{1}_B) = \#\{(x,d) \in G^2 : x, x+d, x+2d \in B\}.$$

There are exactly $|B| = |A|$ trivial contributions, namely $(x, 0)$ with $x \in B$. All other contributions are nonnegative. By Lemma 1.6,

$$A \text{ is 3-AP-free} \iff \Lambda_3(\mathbf{1}_B, \mathbf{1}_B, \mathbf{1}_B) = \frac{|B|}{m^2} = \frac{\delta}{m}. \tag{1.4}$$

We have replaced a question about a set by an identity involving a function.

Remark 1.8 (The whole interval). For $A = [N]$, we have $B = S = \{\overline{n} : n \in [N]\} \subsetneq G$. Then

$$\begin{aligned} m^2 \Lambda_3(\mathbf{1}_S, \mathbf{1}_S, \mathbf{1}_S) &= \#\{(x,d) \in G^2 : x, x+d, x+2d \in S\} \\ &= \#\{(x,y,z) \in S^3 : x + z = 2y \text{ in } G\} \\ &= \#\{(a,b,c) \in [N]^3 : a + c = 2b \text{ in } \mathbb{Z}\} \\ &= \left\lceil \frac{N}{2} \right\rceil^2 + \left\lfloor \frac{N}{2} \right\rfloor^2 \geq \frac{N^2}{2}. \end{aligned} \tag{1.5}$$

The map $(x,d) \mapsto (x, x+d, x+2d)$ is a bijection onto the displayed triples in S^3, with inverse $(x,y,z) \mapsto (x, y-x)$. The equality of the two triple counts follows from Lemma 1.6. Finally, a, c must have the same parity, and each such ordered pair determines $b = (a+c)/2 \in [N]$. This count includes the N constant progressions and both orientations of every nonconstant progression.

Set $Q_N = m^2 \Lambda_3(\mathbf{1}_S, \mathbf{1}_S, \mathbf{1}_S)$ for use below.

1.4 Introducing the instruments

1.4.1 Instrument I: characters

Translation is a basic operation on G. For $d \in G$, put

$$(T_d f)(x) = f(x+d).$$

Which functions change only by a scalar factor when translated?

Definition 1.9. A *character* of G is a group homomorphism $\chi : G \to \mathbb{C}^\times$, where G is additive and $\mathbb{C}^\times$ is multiplicative. Thus $\chi(x+y) = \chi(x)\chi(y)$.

Every character takes values on the unit circle. Indeed, $\chi(x)^m = \chi(mx) = \chi(0) = 1$. For an integer t, define $e_m(\bar{t}) = \exp(2\pi it/m)$. This is well-defined on G, because replacing t by $t + km$ does not change the exponential.

Proposition 1.10 (All characters of the cyclic group)**.** *The characters of G are precisely*

$$\chi_r(x) = e_m(rx), \qquad r \in G.$$

Moreover, $T_d\chi_r = e_m(rd)\chi_r$ for every $d \in G$.

Proof. The expression is independent of the chosen representatives modulo m, and $e_m(r(x+y)) = e_m(rx)e_m(ry)$, so each χ_r is a character. Conversely, a character is determined by $\chi(\bar{1})$. Since $\chi(\bar{1})^m = 1$, it equals $e_m(r)$ for a unique $r \in G$. For an integer representative u of $x \in G$,

$$\chi(x) = \chi(\bar{1})^u = \exp(2\pi i\rho u/m) = e_m(rx),$$

where ρ is an integer representative of r. The formula is independent of both representatives. Finally, $\chi_r(x+d) = e_m(rd)\chi_r(x)$. □

The characters are therefore simultaneous eigenfunctions of all translation operators. They are the basic waves adapted to the cyclic stage.

Lemma 1.11 (Orthogonality)**.** *For $t \in G$,*

$$\mathbb{E}_{x\in G} e_m(tx) = \begin{cases} 1, & t = 0, \\ 0, & t \neq 0. \end{cases}$$

In particular, $\mathbb{E}_x \chi_r(x)\overline{\chi_s(x)} = \mathbf{1}_{r=s}$.

Proof. For $t = 0$, every summand is one. Otherwise, let $z = e_m(t)$. Then $z \neq 1$ and $z^m = 1$, whence

$$\sum_{x=0}^{m-1} z^x = \frac{1-z^m}{1-z} = 0.$$

The last assertion follows by applying the first to $t = r - s$. □

Equip functions on G with the inner product

$$\langle f, g\rangle = \mathbb{E}_x f(x)\overline{g(x)}, \qquad \|f\|_2^2 = \mathbb{E}_x |f(x)|^2.$$

There are m orthonormal characters in the m-dimensional vector space of functions on G. They form an orthonormal basis.

1.4.2 Instrument II: Fourier methods

Definition 1.12 (Fourier transform)**.** For $f : G \to \mathbb{C}$, its Fourier coefficient at frequency r is

$$\widehat{f}(r) = \langle f, \chi_r\rangle = \frac{1}{m}\sum_{x\in G} f(x)e_m(-rx). \tag{1.6}$$

1.4.3 Application of the Fourier transform: counting progressions

Orthogonality gives the identity

$$\mathbf{1}_{\{0\}}(v) = \frac{1}{m}\sum_{r\in G} e_m(rv) \qquad (v \in G).$$

The map $(x,d) \mapsto (x, x+d, x+2d)$ is a bijection from G^2 onto the triples $(x,y,z) \in G^3$ satisfying $x + z = 2y$. Consequently, for functions $f_1, f_2, f_3 : G \to \mathbb{C}$,

$$\begin{aligned}\Lambda_3(f_1,f_2,f_3) &= \frac{1}{m^2}\sum_{\substack{x,y,z\in G\\ x+z=2y}} f_1(x)f_2(y)f_3(z)\\ &= \frac{1}{m^3}\sum_{r\in G}\left(\sum_{x\in G} f_1(x)e_m(rx)\right)\left(\sum_{y\in G} f_2(y)e_m(-2ry)\right)\left(\sum_{z\in G} f_3(z)e_m(rz)\right).\end{aligned}$$

Replace r by $-r$ in the final sum. The definition of the Fourier transform then gives

$$\boxed{\Lambda_3(f_1,f_2,f_3) = \sum_{r\in G}\widehat{f_1}(r)\widehat{f_2}(-2r)\widehat{f_3}(r).} \tag{1.7}$$

In particular, for an integer set $A \subseteq [N]$ and its image $B = \pi(A) \subseteq G$, this identity reads

$$\Lambda_3(\mathbf{1}_B,\mathbf{1}_B,\mathbf{1}_B) = \sum_{r\in G}\widehat{\mathbf{1}_B}(r)^2\widehat{\mathbf{1}_B}(-2r).$$

If A is 3-AP-free, the left-hand side equals $|A|/m^2$ by (1.4). This identity is our starting point for the three taals. We next prove the two Fourier identities used to estimate it.

1.4.4 Fourier inversion

The characters $\chi_r(x) = e_m(rx)$, $r \in G$, form an orthonormal basis for functions on G, with inner product $\langle f, g\rangle = m^{-1}\sum_x f(x)\overline{g(x)}$. The transform gives the coordinates in that basis.

Theorem 1.13 (Fourier inversion)**.** *For every $h : G \to \mathbb{C}$ and every $x \in G$,*

$$h(x) = \sum_{r\in G}\widehat{h}(r)e_m(rx).$$

Proof. Substitute the definition of the transform:

$$\sum_r \widehat{h}(r)e_m(rx) = \frac{1}{m}\sum_{y\in G} h(y)\sum_{r\in G} e_m(r(x-y)).$$

Orthogonality makes the inner sum m if $x = y$, and zero otherwise. Thus the expression equals $h(x)$. □

This is an exact finite sum. There is no convergence question.

1.4.5 Parseval's identity

An orthonormal change of coordinates preserves the squared norm. Here this gives the estimate that will let us control a sum of Fourier products.

Theorem 1.14 (Parseval). *For $f, g : G \to \mathbb{C}$,*

$$\frac{1}{m}\sum_x f(x)\overline{g(x)} = \sum_{r\in G} \widehat{f}(r)\overline{\widehat{g}(r)}.$$

In particular,

$$\sum_{r\in G} |\widehat{f}(r)|^2 = \frac{1}{m}\sum_{x\in G} |f(x)|^2. \tag{1.8}$$

Proof. Expand both functions by inversion. Their inner product becomes

$$\sum_{r,s\in G} \widehat{f}(r)\overline{\widehat{g}(s)}\frac{1}{m}\sum_{x\in G} e_m((r-s)x).$$

The normalized inner sum is 1 when $r = s$, and 0 otherwise. The remaining terms give the first identity. Set $g = f$ for the second. □

For any $C \subseteq G$, this says

$$\widehat{\mathbf{1}_C}(0) = \frac{|C|}{m}, \quad \sum_r |\widehat{\mathbf{1}_C}(r)|^2 = \frac{|C|}{m}, \quad \sum_{r\neq 0} |\widehat{\mathbf{1}_C}(r)|^2 = \frac{|C|}{m} - \left(\frac{|C|}{m}\right)^2. \tag{1.9}$$

The first equality follows directly from the definition, and the last is obtained by removing its squared zero coefficient from the middle sum.

1.4.6 *Comparison with the usual Fourier series*

The closest classical counterpart is Fourier series for 2π-periodic functions. The cyclic group replaces the circle; normalized sums replace normalized integrals.

	Finite cyclic group	Classical circle
Waves	$e^{2\pi irx/m}$, $r \in G$	e^{int}, $n \in \mathbb{Z}$
Coefficients	$\frac{1}{m}\sum_x f(x)e^{-2\pi irx/m}$	$\frac{1}{2\pi}\int_0^{2\pi} F(t)e^{-int}\,dt$
Inversion	$f(x) = \sum_r \widehat{f}(r)e^{2\pi irx/m}$	$F(t) = \sum_n \widehat{F}(n)e^{int}$
Parseval	$\frac{1}{m}\sum_x \lvert f(x)\rvert^2 = \sum_r \lvert\widehat{f}(r)\rvert^2$	$\frac{1}{2\pi}\int_0^{2\pi} \lvert F(t)\rvert^2 dt = \sum_n \lvert\widehat{F}(n)\rvert^2$

In the classical column, for $F \in L^2$, inversion means convergence in L^2; it is not an unconditional claim of pointwise convergence. In the finite column, inversion is an exact finite identity. Orthogonality plays the same role in both theories. In particular, finite Fourier analysis is a change between two coordinate systems: function values and wave coefficients.

1.5 The balanced function

The comparison appropriate to $A \subseteq [N]$ is uniform density Δ on S. Define

$$f_A : G \longrightarrow \mathbb{R}, \qquad f_A = \mathbf{1}_B - \Delta\mathbf{1}_S. \tag{1.10}$$

It equals $1 - \Delta$ on B, $-\Delta$ on $S \setminus B$, and zero outside S. In particular, its sum on G is zero. On the original integer interval,

$$f_A(\pi(n)) = \mathbf{1}_A(n) - \Delta \qquad (n \in [N]).$$

The subscript A records the original integer set; f_A itself is a function on G.

Corollary 1.15. *The balanced function satisfies*

$$\widehat{f_A}(0) = 0, \qquad \widehat{f_A}(r) = \widehat{\mathbf{1}_B}(r) - \Delta\widehat{\mathbf{1}_S}(r),$$

and

$$\sum_{r\neq 0} |\widehat{f_A}(r)|^2 = \frac{N}{m}\Delta(1-\Delta) = \delta(1-\Delta). \tag{1.11}$$

Proof. Its sum is $|B| - \Delta|S| = 0$, proving the first assertion. Linearity gives the second. Finally, its three possible values give

$$\frac{1}{m}\sum_{x\in G} |f_A(x)|^2 = \frac{|B|(1-\Delta)^2 + (|S|-|B|)\Delta^2}{m} = \frac{N}{m}\Delta(1-\Delta).$$

Apply Parseval and remove the zero coefficient, which vanishes. □

Balancing on S removes the Fourier contribution of the ambient interval. For example, if $A = [N]$, then $B = S$, $\Delta = 1$, and $f_A = 0$, although $\widehat{\mathbf{1}_S}(r)$ need not vanish at nonzero frequencies. A large coefficient of f_A detects a deviation from the interval density itself.

1.6 The first taal

Let $N \in \mathbb{N}$, $m = 2N+1$, and $G = \mathbb{Z}/m\mathbb{Z}$. For $A \subseteq [N]$, write

$$\pi(A) = B, \qquad \pi([N]) = S, \qquad \Delta := \frac{|A|}{N} = \frac{|B|}{|S|}, \qquad \delta := \frac{|B|}{m} = \frac{N}{m}\Delta.$$

Keep in mind that Δ is the density on the interval and δ is the density on the group.

Lemma 1.16. *Define*

$$f_A := \mathbf{1}_B - \Delta\mathbf{1}_S, \qquad M := \max_{r\in G\setminus\{0\}} |\widehat{f_A}(r)|.$$

Then

$$\left|\Lambda_3(\mathbf{1}_B, \mathbf{1}_B, \mathbf{1}_B) - \Delta^3\Lambda_3(\mathbf{1}_S, \mathbf{1}_S, \mathbf{1}_S)\right| \leq 3M\delta. \tag{1.12}$$

Proof. By trilinearity, the expression inside the absolute value is

$$\begin{aligned}&\Lambda_3(\mathbf{1}_B, \mathbf{1}_B, \mathbf{1}_B) - \Lambda_3(\Delta\mathbf{1}_S, \Delta\mathbf{1}_S, \Delta\mathbf{1}_S)\\ &\quad = \Lambda_3(f_A, \mathbf{1}_B, \mathbf{1}_B) + \Lambda_3(\Delta\mathbf{1}_S, f_A, \mathbf{1}_B) + \Lambda_3(\Delta\mathbf{1}_S, \Delta\mathbf{1}_S, f_A).\end{aligned}$$

To see the equality, insert and subtract $\Lambda_3(\Delta\mathbf{1}_S, \mathbf{1}_B, \mathbf{1}_B)$ and $\Lambda_3(\Delta\mathbf{1}_S, \Delta\mathbf{1}_S, \mathbf{1}_B)$. Now apply (1.7). The three terms become

$$\begin{aligned}&\sum_{r\in G} \widehat{f_A}(r)\widehat{\mathbf{1}_B}(-2r)\widehat{\mathbf{1}_B}(r),\\ &\sum_{r\in G} \widehat{\Delta\mathbf{1}_S}(r)\widehat{f_A}(-2r)\widehat{\mathbf{1}_B}(r),\\ &\sum_{r\in G} \widehat{\Delta\mathbf{1}_S}(r)\widehat{\Delta\mathbf{1}_S}(-2r)\widehat{f_A}(r).\end{aligned}$$

Since

$$\widehat{f_A}(0) = \frac{1}{m}\sum_{x\in G}\big(\mathbf{1}_B(x) - \Delta\mathbf{1}_S(x)\big) = \frac{|B| - \Delta|S|}{m} = 0,$$

only the summands with $r \neq 0$ survive. Multiplication by -2 permutes G, since m is odd. In particular, $r \neq 0$ implies $-2r \neq 0$. So the f_A factor in each surviving term has absolute value at most M.

For brevity put $u = \Delta \mathbf{1}_S$. By Parseval,

$$\sum_r |\widehat{\mathbf{1}_B}(r)|^2 = \frac{1}{m} \sum_x \mathbf{1}_B(x)^2 = \frac{|B|}{m} = \delta,$$

and

$$\sum_r |\widehat{u}(r)|^2 = \frac{1}{m} \sum_x \Delta^2 \mathbf{1}_S(x)^2 = \frac{\Delta^2 |S|}{m} = \Delta\delta.$$

We use Cauchy–Schwarz in the form

$$\sum_r |a_r b_r| \leq \left(\sum_r |a_r|^2 \right)^{1/2} \left(\sum_r |b_r|^2 \right)^{1/2}.$$

For the first term,

$$\begin{aligned} |\Lambda_3(f_A, \mathbf{1}_B, \mathbf{1}_B)| &\leq M \sum_{r \neq 0} |\widehat{\mathbf{1}_B}(-2r)| |\widehat{\mathbf{1}_B}(r)| \\ &\leq M \left(\sum_r |\widehat{\mathbf{1}_B}(-2r)|^2 \right)^{1/2} \left(\sum_r |\widehat{\mathbf{1}_B}(r)|^2 \right)^{1/2} \\ &= M\sqrt{\delta}\sqrt{\delta} = M\delta. \end{aligned}$$

For the second term,

$$\begin{aligned} |\Lambda_3(u, f_A, \mathbf{1}_B)| &\leq M \sum_{r \neq 0} |\widehat{u}(r)| \, |\widehat{\mathbf{1}_B}(r)| \\ &\leq M \left(\sum_r |\widehat{u}(r)|^2 \right)^{1/2} \left(\sum_r |\widehat{\mathbf{1}_B}(r)|^2 \right)^{1/2} \\ &= M\sqrt{\Delta\delta}\sqrt{\delta} = M\sqrt{\Delta}\,\delta. \end{aligned}$$

For the third term,

$$\begin{aligned} |\Lambda_3(u, u, f_A)| &\leq M \sum_{r \neq 0} |\widehat{u}(r)| \, |\widehat{u}(-2r)| \\ &\leq M \left(\sum_r |\widehat{u}(r)|^2 \right)^{1/2} \left(\sum_r |\widehat{u}(-2r)|^2 \right)^{1/2} \\ &= M\sqrt{\Delta\delta}\sqrt{\Delta\delta} = M\Delta\delta. \end{aligned}$$

In the first and third calculations, the change $r \mapsto -2r$ preserves the sum of squares. Finally, $0 \leq \Delta \leq 1$. Thus

$$\text{L.H.S.} \leq M\delta + M\sqrt{\Delta}\,\delta + M\Delta\delta \leq 3M\delta.$$

□

Theorem 1.17 (The first taal)**.** *If $A \subseteq [N]$ is nonempty and contains no three equally spaced distinct integers, then, whenever $N \geq 4/\Delta^2$,*

$$\max_{r \in G \setminus \{0\}} |\widehat{f_A}(r)| \geq \frac{\Delta^2}{36}. \tag{1.13}$$

Proof. Since A is 3-AP-free,

$$\Lambda_3(\mathbf{1}_B, \mathbf{1}_B, \mathbf{1}_B) = \frac{|B|}{m^2} = \frac{|B|}{|S|}\frac{|S|}{m^2} = \frac{\Delta N}{m^2}.$$

Also, $Q_N = m^2\Lambda_3(\mathbf{1}_S, \mathbf{1}_S, \mathbf{1}_S) \geq N^2/2$. Now

$$N \geq \frac{4}{\Delta^2} \quad \Longrightarrow \quad \frac{\Delta^2 N}{4} \geq 1 \quad \Longrightarrow \quad \frac{\Delta^3 N^2}{4} \geq \Delta N.$$

Therefore

$$\begin{aligned}
&\left|\Lambda_3(\mathbf{1}_B, \mathbf{1}_B, \mathbf{1}_B) - \Delta^3\Lambda_3(\mathbf{1}_S, \mathbf{1}_S, \mathbf{1}_S)\right| \\
&\quad = \frac{1}{m^2}|\Delta N - \Delta^3 Q_N| \\
&\quad = \frac{1}{m^2}(\Delta^3 Q_N - \Delta N) \\
&\quad \geq \frac{1}{m^2}\left(\frac{\Delta^3 N^2}{2} - \frac{\Delta^3 N^2}{4}\right) = \frac{\Delta^3 N^2}{4m^2}.
\end{aligned}$$

The second equality is valid because $\Delta^3 Q_N - \Delta N \geq \Delta^3 N^2/4 > 0$. Thus, by the previous lemma,

$$\frac{\Delta^3 N^2}{4m^2} \leq 3M\delta = 3M\frac{N}{m}\Delta.$$

Since $A \neq \varnothing$, we have $\Delta > 0$, and hence

$$M \geq \frac{m\Delta^3 N^2}{4m^2 \cdot 3N\Delta} = \frac{\Delta^2 N}{12m} \geq \frac{\Delta^2 N}{12(3N)} = \frac{\Delta^2}{36}.$$

Here we used $m = 2N + 1 \leq 3N$. □

Remark 1.18 (The size condition and a fixed density lower bound). The hypothesis $N \geq 4/\Delta^2$ is a condition on the pair (A, N). Since $\Delta = |A|/N > 0$,

$$N \geq \frac{4}{\Delta^2} \quad \Longleftrightarrow \quad |A|^2 \geq 4N \quad \Longleftrightarrow \quad |A| \geq 2\sqrt{N}.$$

It need not become true by enlarging N while keeping $|A|$ fixed. In Roth's theorem, we fix a separate number $\varepsilon > 0$ and assume $\Delta \geq \varepsilon$. Then

$$N \geq \frac{4}{\varepsilon^2} \quad \Longrightarrow \quad N \geq \frac{4}{\Delta^2}.$$

The threshold is therefore uniform over all sets whose interval density is at least that fixed ε.

1.6.1 The coefficient as a sum on the integer interval

Define

$$g : [N] \longrightarrow \mathbb{R}, \qquad g(n) = \mathbf{1}_A(n) - \Delta.$$

The functions g and f_A have different domains. They are related by $f_A(\pi(n)) = g(n)$ for $n \in [N]$. By definition of the Fourier transform,

$$m\widehat{f_A}(r) = \sum_{x \in G} f_A(x)e_m(-rx).$$

Since f_A vanishes outside $S = \pi([N])$, and $\pi : [N] \to S$ is a bijection,

$$m\widehat{f}_A(r) = \sum_{n=1}^{N} f_A(\pi(n)) e_m(-r\pi(n)) = \sum_{n=1}^{N} g(n) e_m(-r\pi(n)).$$

Choose an integer representative ρ of r, and put $\theta = \rho/m$. Then

$$-r\pi(n) = \overline{-\rho n}, \qquad e_m(-r\pi(n)) = e^{-2\pi i \rho n/m} = e^{-2\pi i \theta n}.$$

Thus

$$\boxed{m\widehat{f}_A(r) = \sum_{n=1}^{N} g(n) e^{-2\pi i \theta n}.} \tag{1.14}$$

Replacing ρ by $\rho + km$, $k \in \mathbb{Z}$, multiplies the phase by $e^{-2\pi i k n} = 1$, so the expression is independent of the representative. Also,

$$|g(n)| \le 1, \qquad \sum_{n=1}^{N} g(n) = |A| - \Delta N = 0. \tag{1.15}$$

We now want to turn the large weighted sum supplied by the first taal into a positive unweighted average on an integer progression.

1.7 The second taal: the density increment

Theorem 1.19 (The second taal)**.** *Let $A \subseteq [N]$ be nonempty and 3-AP-free, and let $\Delta = |A|/N$. If*

$$N \ge \left(\frac{7200}{\Delta^2}\right)^2, \tag{1.16}$$

then there is an integer arithmetic progression $P \subseteq [N]$ such that

$$|P| \ge \frac{\Delta^2}{7200}\sqrt{N}, \qquad \frac{|A \cap P|}{|P|} \ge \Delta + \frac{\Delta^2}{108}. \tag{1.17}$$

The progression has a positive integer step and does not wrap around a cyclic group. Its density is measured inside P. We first prove the instruments needed for this conclusion, and then deduce the theorem from (1.13).

1.8 Instruments for the second taal

1.8.1 Item I: pigeonhole approximation

We want a step q for which the phase changes very little when n is replaced by $n + q$.

Lemma 1.20. *For every $\theta \in \mathbb{R}$ and every integer $Q \ge 1$, there are integers q, a such that*

$$1 \le q \le Q, \qquad |q\theta - a| \le \frac{1}{Q}.$$

Proof. Split $[0,1)$ into the Q intervals

$$[0, 1/Q),\ [1/Q, 2/Q),\ \ldots,\ [(Q-1)/Q, 1).$$

Among the $Q+1$ fractional parts $\{j\theta\}$, $0 \le j \le Q$, two belong to the same interval. Choose their indices $j < k$. Write

$$j\theta = b_j + \{j\theta\}, \qquad k\theta = b_k + \{k\theta\}, \qquad b_j, b_k \in \mathbb{Z}.$$

Take $q = k - j$ and $a = b_k - b_j$. Then $1 \le q \le Q$, and

$$|q\theta - a| = |\{k\theta\} - \{j\theta\}| < \frac{1}{Q}.$$

This implies the stated inequality. □

For $n, t \in \mathbb{Z}$, the integer a gives

$$e^{-2\pi i\theta(n+tq)} = e^{-2\pi i\theta n} e^{-2\pi it(q\theta - a)}.$$

Since $|e^{iu} - 1| = 2|\sin(u/2)| \le |u|$, we obtain

$$|e^{-2\pi i\theta(n+tq)} - e^{-2\pi i\theta n}| \le 2\pi|t|\,|q\theta - a| \le \frac{2\pi|t|}{Q}. \tag{1.18}$$

Thus a progression of step q and few terms has nearly constant phase.

1.8.2 Item II: a partition with no discarded points

We will choose the number of terms in each block. The final short part of a residue class will be merged with its preceding block.

Lemma 1.21 (Partition into progression blocks)**.** *Let q, L, N be positive integers with $q \le N$ and $L \le \lfloor N/q \rfloor$. Then $[N]$ can be partitioned into integer progressions $P_1, \dots, P_J$, all of step q, such that*

$$L \le |P_j| < 2L \qquad (1 \le j \le J).$$

Proof. For $1 \le b \le q$, let

$$C_b = \{b, b+q, \dots, b+(M_b-1)q\}, \qquad M_b = \left\lfloor \frac{N-b}{q} \right\rfloor + 1.$$

The sets C_b are disjoint and cover $[N]$. Moreover, $M_b \ge \lfloor N/q \rfloor \ge L$. By division with remainder, write

$$M_b = k_b L + t_b, \qquad k_b \ge 1, \qquad 0 \le t_b < L.$$

Take the first $k_b - 1$ blocks of L successive members of C_b. The remaining $L + t_b$ members form the last block. Every block is a progression of step q, and its number of terms is either L or $L + t_b$, hence lies in $[L, 2L)$. Perform this construction in every residue class. □

The assumption $L \le \lfloor N/q \rfloor$ ensures $k_b \ge 1$ in every residue class. Thus there is always a preceding block into which the remainder can be merged. The values of k_b and t_b may depend on b.

Example 1.22. Take $N = 30$, $q = 3$, and $L = 4$. For each $b = 1, 2, 3$, the class C_b has $M_b = 10 = 2 \cdot 4 + 2$ terms. Thus $k_b = 2$, $t_b = 2$. In each class the first block has $L = 4$ terms and the last has $L + t_b = 6$ terms. Numbering all the blocks together, we get

$$P_1 = \{1, 4, 7, 10\}, \qquad P_2 = \{13, 16, 19, 22, 25, 28\},$$
$$P_3 = \{2, 5, 8, 11\}, \qquad P_4 = \{14, 17, 20, 23, 26, 29\},$$
$$P_5 = \{3, 6, 9, 12\}, \qquad P_6 = \{15, 18, 21, 24, 27, 30\}.$$

Writing $P_j = \{n_j + tq : 0 \le t < \ell_j\}$, the starting points and lengths are

$$(n_1, \dots, n_6) = (1, 13, 2, 14, 3, 15), \qquad (\ell_1, \dots, \ell_6) = (4, 6, 4, 6, 4, 6).$$

So $[30] = P_1 \sqcup \cdots \sqcup P_6$ and $L \le \ell_j < 2L$. Here L is the chosen minimum block length; ℓ_j is the actual number of terms in P_j. The distance between its endpoints is $(\ell_j - 1)q$.

1.8.3 Item III: from a large coefficient to a positive average

The choice of block size has a purpose. With Q of size $\sqrt{N}$, (1.18) bounds the phase variation on a block of fewer than $2L$ terms by $4\pi L/Q$. Since $|g(n)| \le 1$, freezing the phases on all N points then costs at most $4\pi LN/Q$. The original weighted sum will have size at least $2\eta N$. So we choose L of size $\eta\sqrt{N}/100$: the freezing error is a small part of that sum, while the blocks still have a useful lower bound on their size. The following proof checks every bound, including the floor in the definition of L.

Proposition 1.23. *Let $A \subseteq [N]$, $\Delta = |A|/N$, and $f_A = \mathbf{1}_B - \Delta \mathbf{1}_S$. Suppose $0 < \eta \le 1$ and $|\widehat{f_A}(r)| \ge \eta$ for some $r \in G$. If*

$$N \ge (200/\eta)^2,$$

then there is an integer progression $P \subseteq [N]$ such that

$$|P| \ge \frac{\eta}{200}\sqrt{N}, \qquad \frac{|A \cap P|}{|P|} \ge \Delta + \frac{\eta}{3}.$$

Proof. **Step I: choose the step q and the block size L.** Choose an integer representative ρ of r, and put

$$\theta = \rho/m, \qquad Q = \lfloor \sqrt{N} \rfloor, \qquad L = \left\lfloor \frac{\eta\sqrt{N}}{100} \right\rfloor.$$

Here L is a chosen number of terms. We will examine all the blocks in a partition, and then find one with increased density.

The hypothesis gives $\eta\sqrt{N} \ge 200$. Hence $\eta\sqrt{N}/100 \ge 2$. For $x \ge 2$, $\lfloor x \rfloor \ge x/2$. Therefore

$$\frac{\eta\sqrt{N}}{200} \le L \le \frac{\eta\sqrt{N}}{100}. \tag{1.19}$$

Also $N \ge 40000$, so $Q \ge \sqrt{N}/2$. By Lemma 1.20, choose $1 \le q \le Q$ and $a \in \mathbb{Z}$ with $|q\theta - a| \le 1/Q$.

We check that the partition lemma applies. Since $Q^2 \le N$ and $q \le Q$,

$$\left\lfloor \frac{N}{q} \right\rfloor \ge \left\lfloor \frac{N}{Q} \right\rfloor \ge Q.$$

On the other hand,

$$L \le \frac{\eta\sqrt{N}}{100} \le \frac{\sqrt{N}}{100} \le \frac{\sqrt{N}}{2} \le Q.$$

Thus $L \le \lfloor N/q \rfloor$.

Step II: partition $[N]$ and freeze the phases. By Lemma 1.21, write

$$[N] = P_1 \sqcup \cdots \sqcup P_J, \qquad P_j = \{n_j + tq : 0 \le t < \ell_j\}, \qquad L \le \ell_j < 2L.$$

In particular, $\sum_j \ell_j = N$. For $n = n_j + tq \in P_j$, (1.18) gives

$$\begin{aligned} |e^{-2\pi i\theta n} - e^{-2\pi i\theta n_j}| &\le \frac{2\pi t}{Q} \le \frac{4\pi L}{Q} \\ &\le \frac{4\pi(\eta\sqrt{N}/100)}{\sqrt{N}/2} = \frac{8\pi\eta}{100}. \end{aligned} \tag{1.20}$$

Put

$$g(n) = \mathbf{1}_A(n) - \Delta, \qquad s_j = \sum_{n\in P_j} g(n), \qquad c_j = e^{-2\pi i\theta n_j}.$$

Each s_j is real and $|c_j| = 1$. Let $W = \sum_{n=1}^{N} g(n)e^{-2\pi i\theta n}$. Since the blocks partition $[N]$,

$$W - \sum_j c_j s_j = \sum_j \sum_{n\in P_j} g(n)\big(e^{-2\pi i\theta n} - c_j\big).$$

Using $|g(n)| \le 1$ and (1.20),

$$\left|W - \sum_j c_j s_j\right| \le \sum_j \sum_{n\in P_j} \frac{8\pi\eta}{100} = \frac{8\pi\eta N}{100}. \tag{1.21}$$

Step III: obtain a large sum of block discrepancies. By (1.14) and the hypothesis,

$$|W| = m|\widehat{f_A}(r)| \ge m\eta \ge 2\eta N.$$

Therefore

$$\begin{aligned} \sum_j |s_j| &\ge \left|\sum_j c_j s_j\right| \\ &\ge |W| - \left|W - \sum_j c_j s_j\right| \\ &\ge \left(2 - \frac{8\pi}{100}\right)\eta N \ge \eta N. \end{aligned} \tag{1.22}$$

For the last inequality, $8\pi/100 < 1$.

Step IV: choose a block with positive surplus. Again because the blocks cover $[N]$, (1.15) gives

$$\sum_j s_j = \sum_{n=1}^{N} g(n) = 0.$$

Let $J_+ = \{j : s_j > 0\}$. For a real number s, $\max(s, 0) = (|s| + s)/2$. So

$$\sum_{j\in J_+} s_j = \frac{1}{2}\left(\sum_j |s_j| + \sum_j s_j\right) = \frac{1}{2}\sum_j |s_j| \ge \frac{\eta N}{2}.$$

In particular $J_+ \neq \varnothing$. Since $\sum_{j\in J_+} \ell_j \le N$, we have

$$\max_{j\in J_+} \frac{s_j}{\ell_j} \ge \frac{\sum_{j\in J_+} s_j}{\sum_{j\in J_+} \ell_j} \ge \frac{\eta N/2}{N} = \frac{\eta}{2}.$$

The first inequality follows by multiplying every $s_j/\ell_j \le \max_{i\in J_+} s_i/\ell_i$ by ℓ_j and adding. Choose a block $P = P_j$ attaining this maximum. Then

$$\frac{|A\cap P|}{|P|} - \Delta = \frac{1}{|P|}\sum_{n\in P} g(n) = \frac{s_j}{\ell_j} \ge \frac{\eta}{2} \ge \frac{\eta}{3}.$$

Also $|P| = \ell_j \ge L \ge \eta\sqrt{N}/200$, by (1.19). This proves both conclusions. □

The proof gives the stronger gain $\eta/2$. The weaker bound $\eta/3$ stated in the proposition is sufficient for the second taal.

1.9 Proof of the second taal

Proof of Theorem 1.19. Since $0 < \Delta \le 1$, the size hypothesis gives

$$N \ge \frac{7200^2}{\Delta^4} \ge \frac{4}{\Delta^2}.$$

So the first taal supplies a nonzero r for which $|\widehat{f_A}(r)| \ge \Delta^2/36$. Put $\eta = \Delta^2/36$. Then $0 < \eta \le 1$, and

$$(200/\eta)^2 = \left(\frac{7200}{\Delta^2}\right)^2 \le N.$$

Apply Proposition 1.23. Its two conclusions become

$$|P| \ge \frac{\eta\sqrt{N}}{200} = \frac{\Delta^2\sqrt{N}}{7200}, \qquad \frac{|A \cap P|}{|P|} \ge \Delta + \frac{\eta}{3} = \Delta + \frac{\Delta^2}{108}.$$

These are exactly (1.17). □

1.9.1 Relabel the progression and start again

Write $P = \{a + qt : 0 \le t < \ell\} \subseteq [N]$, where $q \ge 1$. Define

$$A_P := \{t + 1 \in [\ell] : a + qt \in A\}.$$

The map $u \mapsto a + q(u-1)$ is a bijection from $[\ell]$ to P. Thus

$$|A_P| = |A \cap P|, \qquad \frac{|A_P|}{\ell} = \frac{|A \cap P|}{|P|}.$$

For $u_1, u_2, u_3 \in [\ell]$,

$$\begin{aligned}[a + q(u_1 - 1)] + [a + q(u_3 - 1)] - 2[a + q(u_2 - 1)] \\ = q(u_1 + u_3 - 2u_2).\end{aligned}$$

Since $q \ne 0$, the images satisfy the progression equation if and only if the u_i do. The map is injective, so it also preserves distinctness. Hence A_P is 3-AP-free whenever A is 3-AP-free.

Put $N' = \ell$, $A' = A_P$, and $\Delta' = |A'|/N'$. We have proved

$$\Delta' \ge \Delta + \frac{\Delta^2}{108}, \qquad N' \ge \frac{\Delta^2}{7200}\sqrt{N}. \tag{1.23}$$

The next application uses $G' = \mathbb{Z}/(2N'+1)\mathbb{Z}$, together with the new image B', interval image S', and group density $\delta' = N'\Delta'/(2N'+1)$. Thus the interval density at every stage continues to be denoted by a capital Δ.

Remark 1.24 (What changes at each step?). At stage j, the construction gives an integer set $A_j \subseteq [N_j]$, a progression $P_j \subseteq [N_j]$, and a relabelled set $A_{j+1} \subseteq [N_{j+1}]$, where $N_{j+1} = |P_j|$. Thus

$$|A_{j+1}| = |A_j \cap P_j| \le |A_j|, \qquad N_{j+1} \le N_j, \qquad \Delta_{j+1} \ge \Delta_j + \frac{\Delta_j^2}{108}.$$

The sets contain no more elements, while their relative densities rise. In fact $N_{j+1} < N_j$: equality would force $P_j = [N_j]$, whose density is exactly Δ_j, contradicting the positive increment. After relabelling, the sets need not be nested as subsets of $\mathbb{Z}$. The inclusion is $A_j \cap P_j \subseteq A_j$ before that relabelling.

1.10 Instruments for the third taal

The density rises at each step, while the interval length decreases. The remaining issue is whether the interval stays large enough for the next application. We will fix an initial density lower bound ε, determine how many steps suffice, and then choose the starting length backwards.

Lemma 1.25 (Keeping enough length for finitely many steps)**.** *Fix $0 < \varepsilon \le 1$, and let K be a positive integer. Set*

$$R_0 = \left\lceil \left(\frac{7200}{\varepsilon^2}\right)^2 \right\rceil, \qquad R_{t+1} = \left\lceil \left(\frac{7200 R_t}{\varepsilon^2}\right)^2 \right\rceil \quad (0 \le t < K).$$

Suppose $0 \le s \le K$, and positive integers $N_0, \dots, N_s$ satisfy

$$N_0 \ge R_K, \qquad N_{j+1} \ge \frac{\varepsilon^2}{7200}\sqrt{N_j} \quad (0 \le j < s).$$

Then $N_j \ge R_{K-j} \ge R_0$ for $0 \le j \le s$.

Proof. Every R_t is a positive integer. Also $7200/\varepsilon^2 \ge 1$, so $R_{t+1} \ge R_t$. By its definition,

$$R_{t+1} \ge \left(\frac{7200 R_t}{\varepsilon^2}\right)^2 \quad \Longrightarrow \quad \frac{\varepsilon^2}{7200}\sqrt{R_{t+1}} \ge R_t.$$

We prove $N_j \ge R_{K-j}$ by induction. At $j = 0$, this is the assumption. If $j < s$ and the assertion holds at j, then

$$N_{j+1} \ge \frac{\varepsilon^2}{7200}\sqrt{N_j} \ge \frac{\varepsilon^2}{7200}\sqrt{R_{K-j}} \ge R_{K-j-1}.$$

Here $K - j \ge 1$, so the last comparison is one of the displayed inequalities above. This completes the induction. Monotonicity of the R_t gives $R_{K-j} \ge R_0$. □

1.11 The third taal

Proposition 1.26 (Finite iteration)**.** *Fix $0 < \varepsilon \le 1$, and put*

$$K = \left\lceil \frac{108}{\varepsilon^2} \right\rceil, \qquad N_*(\varepsilon) = R_K,$$

with the R_t from Lemma 1.25. Starting from a 3-AP-free set $A_0 \subseteq [N_0]$ with $N_0 \ge N_*(\varepsilon)$ and $\Delta_0 = |A_0|/N_0 \ge \varepsilon$, the second taal can be applied at every stage $j < K$ that is reached. Continuing through those stages would force a density greater than one. Consequently, such an initial set cannot exist.

Proof. Suppose such an A_0 exists. After j steps, write $A_j \subseteq [N_j]$ and $\Delta_j = |A_j|/N_j$. We claim that, at each stage $0 \le j \le K$ reached by the construction,

$$A_j \text{ is 3-AP-free}, \quad N_j \ge R_{K-j}, \quad \Delta_j \ge \varepsilon + \frac{j\varepsilon^2}{108}. \tag{1.24}$$

At $j = 0$, these follow from the assumptions. Suppose they hold at $j < K$. Since $A_j \subseteq [N_j]$, we have $\Delta_j \le 1$. The lower bound in (1.24) also gives $\Delta_j \ge \varepsilon > 0$, so $A_j \ne \varnothing$. Moreover,

$$N_j \ge R_{K-j} \ge R_0 \ge \left(\frac{7200}{\varepsilon^2}\right)^2 \ge \left(\frac{7200}{\Delta_j^2}\right)^2.$$

Every hypothesis of the second taal is therefore satisfied. It gives a progression $P_j \subseteq [N_j]$. Relabel $A_j \cap P_j$ as $A_{j+1} \subseteq [N_{j+1}]$, where $N_{j+1} = |P_j|$. By the relabelling argument, the new set is again 3-AP-free. By (1.23),

$$\Delta_{j+1} \geq \Delta_j + \frac{\Delta_j^2}{108} \geq \Delta_j + \frac{\varepsilon^2}{108} \geq \varepsilon + \frac{(j+1)\varepsilon^2}{108},$$
$$N_{j+1} \geq \frac{\Delta_j^2}{7200}\sqrt{N_j} \geq \frac{\varepsilon^2}{7200}\sqrt{R_{K-j}} \geq R_{K-j-1}.$$

Thus all assertions propagate to the next stage. This proves both that the next application is justified and that the claimed bounds hold.

After K steps we obtain

$$\Delta_K \geq \varepsilon + \frac{K\varepsilon^2}{108} \geq \varepsilon + 1 > 1.$$

But $\Delta_K = |A_K|/N_K \leq 1$, a contradiction. □

The number of steps K and the starting threshold $N_*(\varepsilon)$ depend only on the fixed positive lower bound ε. The actual densities Δ_j are free to change. The auxiliary thresholds R_t increase with t, whereas the bound used at stage j is R_{K-j}, which decreases as j increases.

1.12 Combining the three taals: Roth's theorem

Theorem 1.27 (Roth)**.** *For every $\varepsilon > 0$, there is an integer $N_0(\varepsilon)$ such that, whenever $N \geq N_0(\varepsilon)$ and $A \subseteq [N]$ satisfies $|A| \geq \varepsilon N$, the set A contains three equally spaced distinct integers.*

Proof. If $\varepsilon > 1$, there are no sets satisfying the size hypothesis, so the assertion is vacuous. Suppose $0 < \varepsilon \leq 1$, and take $N_0(\varepsilon) = N_*(\varepsilon)$ from the third taal. If $A \subseteq [N]$ satisfied the size hypothesis and were 3-AP-free, it would be an initial set forbidden by Proposition 1.26. So A must contain a nontrivial three-term progression. □

Corollary 1.28. *Let $r_3(N) = \max\{|A| : A \subseteq [N],\ A \text{ is 3-AP-free}\}$. Then $r_3(N) = o(N)$.*

Proof. Fix $\varepsilon > 0$. For $N \geq N_0(\varepsilon)$, Roth's theorem says that every 3-AP-free $A \subseteq [N]$ has $|A| < \varepsilon N$. Therefore $0 \leq r_3(N)/N < \varepsilon$ for all sufficiently large N. Since ε was arbitrary, the ratio tends to zero. □

Corollary 1.29. *Every 3-AP-free set $D \subseteq \mathbb{N}$ has natural density zero. Equivalently, every subset of $\mathbb{N}$ of positive upper density contains a nontrivial three-term progression.*

Proof. If D is 3-AP-free, then $D \cap [N]$ is 3-AP-free for every N. Hence

$$0 \leq \frac{|D \cap [N]|}{N} \leq \frac{r_3(N)}{N} \longrightarrow 0.$$

Thus the natural density exists and equals zero. For the equivalent formulation, if $\limsup_{N\to\infty} |D \cap [N]|/N > 0$, choose $\varepsilon > 0$ smaller than this upper density. There are arbitrarily large N with $|D \cap [N]| \geq \varepsilon N$. Choose one also satisfying $N \geq N_0(\varepsilon)$, and apply Roth's theorem to $D \cap [N]$. □

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